Dipon Deb Dipu

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M.Sc. in Embedded Systems
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Professional Summary

Independent and solutions-driven Master's student in Embedded Systems at Uppsala University with a strong foundation in hardware/software co-design, artificial intelligence, and digital logic. Experienced in developing deep learning models using PyTorch and designing digital hardware architectures. Actively expanding expertise in FPGA programming and model-based design to bridge the gap between trained AI models and synthesizable RTL for efficient hardware acceleration. Available for full-time thesis work.

Education

- **M.Sc. in Embedded Systems** *Sweden*
  Uppsala University 2025 – 2027
  Full Scholarship Recipient | Focus: FPGA, Arch.

- **B.Sc. in CSE** *Bangladesh*
  Ahsanullah Univ. of Sci. & Tech. 2020 – 2024
  CGPA: 3.75 / 4.00 | Dean's List of Honor

Job Experience

- **Backend Developer & Sys Admin** *Dhaka, Bangladesh*
  Arthur Private Company Ltd. Feb 2023 – Jun 2025
  - Administered backend systems, scalable architectures, and technical configurations with high precision.
  - Developed automation scripts and CI workflows to streamline hardware/software testing pipelines.

Key Skills & Technical Stack

AI & ML: PyTorch, Deep Learning, CNNs, Computer Vision, GANs.

Hardware & RTL: FPGA Programming, RTL Concepts, Digital Logic, Microprocessors.

Programming: Python, C/C++, SystemVerilog, Assembly (8086), MATLAB.

Simulation & EDA: AMD Vivado, gem5, Simulink, Proteus 8, Webots.

Systems & Networking: Git/GitHub, CI/CD, Embedded Linux.

Leadership & Event Logistics

- **General Secretary & Vice President** *Bangladesh*
  AUST CSE Society & Research Club 2022 – 2024
  - Directed large-scale showcases, managing physical packing, transport, and setup of tech hardware.
  - Co-founded the Research Club and organized the 2023 Research Symposium for students and faculty.

Languages

English (Fluent/Professional)

Bengali (Native)

Projects & Publications

- **FPGA Programming & Model-Based Design** *2026*
  AMD Vivado, MATLAB, Simulink
  - Developing and synthesizing digital designs targeting FPGA hardware utilizing AMD Vivado.
  - Implementing model-based design workflows via MATLAB/Simulink to simulate physical systems and generate embedded code.

- **Peer-Reviewed IEEE Publications** *2024*
  PyTorch, Deep Learning, Medical Imaging
  - Authored **"NeuroMindscape: Deep Learning Unraveling Alzheimer's Stages"** (IEEE: 10800075) proposing a 7-layer CNN.
  - Co-authored **"A Hybrid Classical-Quantum Deep Learning Framework"** (IEEE: 11252366), leveraging PennyLane and PyTorch for a 96.26% accuracy QML model.

- **Computer Architecture Simulation** *2026*
  gem5 Simulator, Linux/WSL2, HW/SW Co-design
  - Installed and configured the gem5 system simulator within a Linux environment to explore computer architecture and hardware/software limits.
  - Compiled gem5 targets, executed simulation scripts, and analyzed system performance for cross-compilation.

- **Digital Hardware Architecture Simulation** *2024*
  Proteus 8, 8086 Assembly, Digital Logic Design
  - Designed and implemented digital hardware architectures including ALUs, Booth multipliers, and SAP computers.
  - Conducted rigorous simulation and testing of digital circuits using Proteus 8 prior to physical deployment.
  - Programmed natively in 8086 Assembly, analyzing instruction sets, cycle times, and memory interfaces.

- **Wireless Backscatter & Network Consensus** *2025*
  C/C++, nRF52840, CC1352P7, Raspberry Pi
  - Engineered a wireless embedded systems protocol optimizing communication utilizing backscatter tags.
  - Managed physical layer packet measurements, focusing on low-power RF signaling and tuning logic.